

Double-Tonic Complexes in Rock Music

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Most analysts consider a song either to be in a single key or to exhibit competition or ambiguity among multiple possible keys. This article proposes an alternative in which two keys combine to create a coherent, stable tonality. I adapt Robert Bailey's concept of the "double-tonic complex" to demonstrate that in many rock songs, relative major and minor keys coexist with neither superior to the other and no conflict between them. These songs are not quite in two keys at once, but rather each component key represents a different incarnation of a single, more abstract tonality encompassing them both. The prolonged generative tonic of such a structure is a four-note sonority built from the union of the two tonic triads; this sonority often—but not always—appears as a surface chord at structurally important moments.

Keywords: rock music, popular music, tonality, harmony, double-tonic complex.

Many rock songs have a puzzling feature: despite ostensibly simple harmonic structures, it is not entirely clear what key they are in. Indeed, it is perhaps *because* of rock's harmonic simplicity that an unequivocal key often fails to emerge. Recent works exploring unclear tonality in rock music tend either to argue for why one option is the song's *true key*, or to claim *ambiguity* or *competition* among the possible key centers.¹ Both arguments stem from an underlying assumption that keys are mutually incompatible—that if no single center asserts itself unequivocally, then there must be some sort of tonal struggle. Christopher Doll, whose 2017 monograph devotes an entire chapter to ambiguity, describes competing tonal centers as follows:

Our hearing two simultaneous centers is like facing two different directions at once, or . . . involving a flickering or oscillating quality generated by two or more *competing and mutually exclusive* interpretations. . . . [In an ambiguous two-chord loop,] schemas and functions are not simply multiple, they are hierarchically contradictory: if the first chord is a tonic and resides at a deeper harmonic level, the same cannot also be true of the second chord.²

To be sure, the word "ambiguity" in its strictest sense signifies multiple distinct possible interpretations. In this article, I investigate an alternative to the ambiguity paradigm: a tonal situation that exhibits neither one single key nor competition among multiple keys.

Music theorists have brought up this possibility before. The intricate tonal practices of late nineteenth-century Austro-German composers led theorists to posit "alternatives to monotonicity," applying to tonal pieces that do not coalesce around a single global tonic. Much of this research derives from the work of Robert Bailey, particularly through the use of

the Baileyan concepts of *tonal pairing*, *directional* or *progressive tonality*, and the *double-tonic complex*.³ Tonal pairing exists when two keys are presented in juxtaposition or alternation with neither obviously superior to the other. Directional/progressive tonality describes a transition from one key to a different one across an entire piece. And the double-tonic complex, a more controversial concept, arises when two keys a third apart combine to create a controlling tonality. The keys are not in conflict, nor is there an ambiguity as to which is primary; as Bailey explains, "these two tonalities are not really set in opposition to each other like the contrasting keys found in earlier practice; rather, they are co-existent."⁴ As you have likely surmised from my title, this article focuses on this last concept.

THE DOUBLE-TONIC COMPLEX

The double-tonic complex is a special type of tonality where two keys coexist. It is not quite a matter of hearing two keys at once, though. Rather, each component key represents a different incarnation of a single, more abstract tonality encompassing them both. The concept of the double-tonic complex as an abstract tonality encompassing two keys distinguishes it from tonal pairing. In the latter, the two keys are incompatible, and the ambiguity between them can become a source of tonal conflict. In other words, tonal pairing is not itself a tonal system but rather represents a juxtaposition of two monotonal frameworks. An analogy to the ancient Greek concept of concordant versus discordant intervals might clarify the distinction: two notes forming a concordant interval combine to create a single perceptual entity (a consonance), while two notes forming a discordant interval remain separate notes

¹ See, for instance, Schultz (2012), Rusch (2013), Osborn (2016), Doll (2017), Richards (2017a), and Spicer (2017).

² Doll (2017, 219), emphasis mine.

³ See Bailey (1969, 1977, 1978, 1985) and Krebs (1981), Lewis (1984), Stein (1985), Kinderman (1988), and many essays in Kinderman and Krebs (1996), among others.

⁴ Bailey (1978), as quoted in Lewis (1984, 4).

(a)

(b)

EXAMPLE 1. Nineteenth-century examples of four-note sonorities functioning as tonic within a double-tonic complex. Both are built from the union of A-minor and C-major triads

(i.e., there is no such *thing* as a dissonance).⁵ We could say that in a double-tonic complex, the two component keys are consonant with one another, whereas in tonal pairing, they are dissonant.

Bailey's original formulation of the double-tonic complex is opaque, and not all theorists have employed the concept in the same way.⁶ Matt BaileyShea divides the various interpretations of the double-tonic complex into three categories: tonal pairing (treating the two as synonyms), "motivic" double-tonic complex (mostly the same as tonal pairing but with some sort of motivic intertwining of the two keys), and "structural" double-tonic complex (where the conflation of two keys acts as the piece's controlling tonality).⁷ A structural double-tonic complex involves a four-note sonority, built from the union of two third-related triads, acting as a prolonged tonic—for example, A–C–E–G as the union of A-minor and C-major triads, which occurs at climactic moments in both Wagner's *Tristan und Isolde* and Mahler's *Das Lied von der Erde*, as shown in Example 1. Bailey seems to aim toward a structural conception of the double-tonic complex in his theoretical explanations, but rarely provides thorough analytical support for such a structure.⁸ BaileyShea even posits that structural double-tonic

complexes are "arguably impossible in nineteenth-century music."⁹ Daniel Harrison agrees with BaileyShea in the case of Wagner ("Wagner's art did not recognize a fusion of the two triads ... as a possible tonic sonority") but argues that such fused tonics are characteristic of French Impressionism and tonal practices in the early to mid-twentieth century.¹⁰

In my view, the four-note tonic sonority is fundamental to the idea of the double-tonic complex as a specific type of tonality. The four-note sonority embodies the fusion of two keys into a broader framework; in a structural double-tonic complex, this sonority is not merely a surface phenomenon, but represents the tonal foundation underlying a passage or entire work. While evidence for such prolonged sonorities is limited in the nineteenth-century literature, rock music's particular harmonic idiom can provide an especially ripe environment for structural double-tonic complexes. Two features in particular make this so: rock's frequent employment of seventh chords as stable sonorities, and its common use of Aeolian modality. The former means that perceiving the four-note union of two third-related triads as a consonance is not problematic.

⁸ See, e.g., Bailey (1985, 121–22).

⁹ BaileyShea (2007, 195).

¹⁰ Harrison (2016, 121–24). Harrison cites Pomeroy (2004) (on Debussy) and Straus (1982) (on Stravinsky) as representative studies supporting fused chords as tonic sonorities.

⁵ See Barker (1989, 193n7).

⁶ A good summary of Bailey's formulation appears in Lewis (1984, 4–6).

⁷ BaileyShea (2007).

at 1:13

C# G# F#

tonic sonority is
union of A#m
and C#M triads

Everything means nothing to me, Everything means nothing to me

melody
prolongs A#

EXAMPLE 2. Elliott Smith, “Everything Means Nothing to Me” (2000): a four-note sonority built from the union of two triads functions as prolonged tonic (after [Schultz \[2012\]](#))

The latter means that the distinction between two relative keys is often negligible beyond their different tonic chords.

Some theorists have brought up the possibility of double-tonic complexes in rock music. Most of their analyses reside squarely in BaileyShea’s first two categories—i.e., tonal pairing with or without some motivic intertwining—including René Rusch’s analysis of Radiohead’s “Paranoid Android,”¹¹ Brad Osborn’s analysis of the same band’s “Reckoner,”¹² and both Robert Gauldin’s and Walter Everett’s analyses of the Beatles’ *Abbey Road* medley (both of which I will discuss alongside my own analysis of the medley below).¹³ One analysis that approaches a structural interpretation of the double-tonic complex comes from Rob Schultz, who, ironically, uses the term “tonal pairing” instead of “double-tonic complex.” Schultz’s analysis of Elliott Smith’s “Everything Means Nothing to Me,” summarized in [Example 2](#), convincingly demonstrates that the keys of A# minor and C# major fuse into a prolonged four-note tonic sonority governing the song’s extended outro. Schultz underscores this sonority’s status as a fused tonic rather than a dissonant chord: “although it resembles a C#-add-6 chord in construction, the governing tonal context instead favors its being construed simply as the union of an A# minor and C# major triad.”¹⁴

TONIC AND KEY

Before proceeding to more detailed analyses, I would like to clarify my use of the terms “tonic” and “key” in this article. In short, “tonic” refers to a harmonic entity, while “key” refers to a broader system of tonal relationships. A key is specifically defined in reference to a *tonal center*—a single, maximally stable pitch class—and can be further distinguished as either major or minor, based on whether the prevailing diatonic collection

includes a major or minor third above the tonal center (e.g., the key of C major, or the key of A minor). Some theorists question the validity of the major/minor binary in reference to rock due to its use of various diatonic modes as referential pitch collections; without veering too far from this article’s main focus, I will say that while it is true that some rock songs do not fit neatly into either the major or minor category (as is the case in many classical works), most songs consistently use either $\hat{3}$ or $b\hat{3}$ in their harmonic content, especially in chords built upon their tonal center (i.e., I chords). In other words, I consider Ionian and Mixolydian to be diatonic collections available in a major context (i.e., $\hat{7}$ has two valid versions), and Dorian and Aeolian (as well as harmonic and melodic minor) to be pitch collections available in a minor context (i.e., both $\hat{6}$ and $\hat{7}$ have two valid versions).¹⁵

When describing a particular chord within a given song, “tonic” identifies it as “a harmony stable enough to preclude any sense of requiring resolution,” to use Doll’s words.¹⁶ There may be multiple possible combinations of pitch classes that could be called tonics at any given moment; for example, EM, Em, E5, and a unison E could all be tonic harmonies with a tonal center of E. More abstractly, though, “tonic” refers to a harmonic entity that in some way *generates* the pitch relationships governing a stretch of music. Here we may speak not of *a* tonic but rather *the* tonic. Conceiving of tonic as a generative harmonic entity echoes the Schenkerian view that all tonal relationships arise organically from the “chord of nature” (i.e., the major triad); as Christopher Lewis puts it, “a common-practice piece . . . is not simply ‘in’ a given key, but expresses a *single tonic triad*.”¹⁷ In the key of E major, we might say that the triad E–G#–B is the generative tonic, and all chords and melodic pitches are defined in reference to that sonority. Further, a given song may contain no tonic chords at all but nevertheless revolve around an unstated generative tonic,

¹¹ Rusch (2013, esp. [2.3]–[2.9]).

¹² Osborn (2016, 149–51, cf. 170n23).

¹³ Gauldin (1990) and Everett (1995; 1999, 269–71). See also Guy Capuzzo’s (2009) concept of “sectional tonality,” where two or more keys combine to form a governing tonality. In contrast to the double-tonic complex, however, sectional tonality involves keys that are temporally separate, each governing one or more distinct song sections without interacting at any local level.

¹⁴ Schultz (2012, 22–30).

¹⁵ David Temperley’s statistical analysis supports both the notion that rock songs rarely mix $\hat{3}$ and $b\hat{3}$ within the harmonic domain and that $\hat{7}$ frequently appears in two forms (as does $\hat{6}$ in minor), even though Temperley himself eschews the major/minor labeling system; see [Temperley \(2018, Example 2.12\[b\]\)](#).

¹⁶ Doll (2017, 20).

¹⁷ Lewis (1984, 2); see also Schenker ([1935] 1979, 11), and Schachter (1995).

VERSE Verse solidly establishes C major; most chords are triads

C A $\flat$ maj7 B $\flat$

10
8 I see you, and you see me watch you blowin' the lines, when you're making a scene, oh girl,
you play with love you can twist it around, baby that ain't enough, cause girl,

PRECHORUS

C A $\flat$ maj7 B $\flat$ C F/D

14
8 you got to know what my head overlooks the senses will show to my heart when it's
I'm gon-na know if you're letting me in or letting me go, don't lie when you're

Prechorus introduces chord structure of a major triad plus a different bass note

CHORUS Upper voices project C major; bass is centered on A

C/A G/E A $\flat$ F/G G/A C/A G/E C/A

19
8 watching for lies, well } you can't escape my Pri - vate eyes, they're watching you, they see your
hurting inside, 'cause

Chorus begins with A-C-E-G sonority

EXAMPLE 3. Daryl Hall and John Oates, “Private Eyes” (1981): simplified transcription of verse, prechorus, and chorus

reflecting what Mark Spicer calls an “absent tonic.”¹⁸ At issue here is whether a tonic requires a tonal center. The two concepts are usually inextricably linked, as any tonic harmony is generally understood to be built upon the tonal center. The double-tonic complex, though, proposes a generative tonic entity that does not rely on a single tonal center. In the A/C complex, for instance, the generative tonic is the four-note chord A–C–E–G (as a union of two triads, *not* an A $\flat$ m⁷ chord; it could also manifest as C–E–G–A, resembling a C⁶ chord), and both A-minor and C-major triads can act as tonic harmonies. We cannot quite say that we are in the key of either A minor or C major, nor are we in both keys at once; rather, we find ourselves in a tonal system based on the relationships

common to both A minor and C major without specifying one as structurally superior to the other.

DARYL HALL AND JOHN OATES, “PRIVATE EYES”

To see how this plays out in the rock repertoire, let’s turn to Daryl Hall and John Oates’s 1981 hit “Private Eyes.” Example 3 gives a transcription of the main verse–prechorus–chorus cycle, with the vocal parts on the top staff, a rhythmically simplified version of the piano ostinato on the middle staff, and a reduced bass line on the bottom (I will explain the idiosyncratic chord labels a bit later). At first glance, it seems that we can divide this song’s sections into distinct keys: the verses are in C major and the choruses are in A minor, with the prechorus acting as a transition between the two. At a basic level, then,

¹⁸ Spicer (2017).

The musical score is divided into two systems. The first system (measures 24-28) features a vocal melody in treble clef and piano accompaniment in grand staff. Chords are labeled above the staff: C/D, F/D, F/G, G/A, C/A, G/E, C/A, C/D, F/D. The lyrics are: "e - v'ry move. Pri - vate eyes, they're watching you. Pri - vate eyes". The second system (measures 29-32) continues the vocal melody and piano accompaniment. Chords are labeled: A♭/B♭, C/A, G. The lyrics are: "watching you, watching you, watching you, watching you" and "You play with words". A box highlights a semitonal voice leading in the piano part: A♭ → G, E♭ → E♮, C → C, B♭ → A. A note states: "Semitonal voice leading cadencing into the four-note tonic sonority". Another box notes: "This riff continues into the next verse, but the bass note changes from A to C".

EXAMPLE 3. [Continued]

we have a juxtaposition of two relative keys in succession, potentially displaying tonal pairing or even Guy Capuzzo's "sectional tonality."¹⁹ But as we look closer we see that A minor and C major become more and more intertwined throughout the song.

The song begins with an intro based on the chorus's chord progression (not shown in Example 3), which in isolation seems to point toward the key of A minor. The ensuing verse, by contrast, is solidly centered on C. In the verse, a mixture-laden I–♭VI–♭VII progression accompanies a vocal melody embellishing an arpeggiation of the C-major triad, focusing on the note E.²⁰ The verse is the only section of the song in which the majority of chords are simple triads. Beginning in the prechorus, most chords contain four notes, with the piano's right hand playing a major triad over a bass note that is not part of that triad. The two resulting sonorities are minor-seventh chords, with a major triad built a minor third above the bass note, and what Mark Spicer has called "soul dominants," with a major triad built a minor seventh above the bass note. Trevor de Clercq has suggested that such four-note chords might exhibit a "harmonic-bass divorce," where the bass and the other instrumental parts operate somewhat independently.²¹ In de Clercq's interpretation, many of rock

music's chord extensions—even simple seventh chords—arise from the combination of a triad with an independent bass note. I believe "Private Eyes" is a prime example of a harmonic-bass divorce; for this reason, I have labeled all chords with slash notation, even those chords that could otherwise be analyzed as minor-seventh chords. So, C/A indicates a C-major triad with an A bass note, F/G indicates an F-major triad with a G bass note, and so on. The piano/bass stratification continues through the chorus and eventually the bridge.

The progression in the prechorus does not contradict the verse's C-centricity, though it could also function within an A-minor context. The section exhibits the typical prechorus outline of pre-dominant to dominant in either key, whether we follow the bass or the piano (or both). Once we get to the chorus, the focus seems to shift away from C toward A. If we look closer, though, we see that C major has not entirely disappeared. In fact, if we take out the bass line, what remains is solidly grounded in C major, essentially looping a I–IV–V–I progression. Given the harmonic stratification between piano and bass, it might not be such a stretch to hear the chorus's opening and closing C/A chords not as Am⁷ chords but instead as consonant unions of A-minor and C-major triads. The transition from the end of the chorus to the second verse guides us further toward hearing this triadic union. In m. 31, the chorus cadences to the A–C–E–G sonority. In mm. 31–32, the piano embellishes a C-major triad with a riff in eighth

¹⁹ Capuzzo (2009).

²⁰ As described above, I consider ♭VII to be endemic to the major mode as a Mixolydian option; ♭VI, however, is borrowed from minor and thus exhibits modal mixture. See also Biamonte (2010, 101).

²¹ de Clercq (2019).

BRIDGE
starts at 2:01

69 F/D C/A F/D C/A B $\flat$ /G

Ooh, why you try to put up a front for me? I'm a spy, but on your side, you see. Slip on into

74 E7

a - ny disguise, I'll still know you. Look in - to my pri - vate eyes, they're watching you

CHORUS

EXAMPLE 4. Daryl Hall and John Oates, "Private Eyes," bridge

Intro Verse Prechorus Chorus Bridge Chorus

a: i $\flat$ III iv 7 $\flat$ VII 9_4 i iv 9 V 7 i

EXAMPLE 5. Possible monotonal reading of "Private Eyes" in *A* minor

notes over the bass note A (the transcription simplifies the rhythm to long notes). After two measures, we return to the verse, where the same piano riff continues (mm. 10–11), but now over the bass note C! The verse and chorus are further linked by opening with the same anacrustic melodic motive, harmonized with C major in the former ("I see you and you see me") and A–C–E–G in the latter ("Private eyes, they're watching you"). The sections' similarities suggest two things: (1) C–E–G in the piano's right hand represents a C-major triad, not the third, fifth, and seventh of an Am 7 chord, even when the bass is on A; and (2) the verses' C-major triads are *incomplete* versions of a four-note tonic sonority, the full version of which is stated in mm. 31–32. They are not incomplete in the sense of a Riemannian *Scheinkonsonanz*, where an unstated A is implied, or even perceived at some abstract level.²² Rather, the C-major triads represent just one side of the global tonic, the yin to A minor's yang, neither of which tells the whole story without reference to the other.

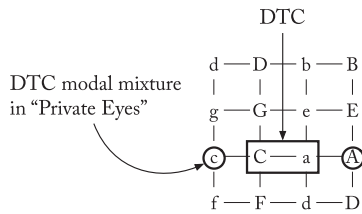
²² Riemann (1890, Chapter 2); see also Kopp (2002, 85–86).

Many theories of rock tonality consider a song's key to be "major until proven otherwise," as Osborn describes it.²³ So why don't we simply say that this song is in C major? We certainly could leave it at that, but there remains something unsatisfying about that interpretation; despite the verses' clear C-major tonality, the song really seems to place its greatest structural emphasis away from C. The song's first and last sections (the intro and chorus) are geared more toward A, and the strongest points of arrival occur on the A–C–E–G sonorities at the beginning and end of the chorus.²⁴ Further, the bridge, shown in Example 4, shines a spotlight on A minor by ending on an E 7 chord, A minor's true tonal dominant, leading tone and all. As Osborn has argued, the tonal dominant-tonic axis can often trump all other factors in pointing toward a specific tonal center.²⁵ Taking a look at the deep-level tonal structure, it is not difficult to forge a monotonal reading in A minor such as that shown in Example 5, where the verses' C major represents A minor's upper third.

²³ Osborn (2016, 150); see also Richards (2017a, 10–11) on the "major-mode bias."

²⁴ Despite the piano/bass stratification set up in the prechorus and the motivic and accompanimental links between verse and chorus, the chorus's opening seems like a relatively solid arrival on A minor. In addition to the placement of A in the bass, the G in the piano part is not even particularly audible in this chord—the pianist might just be playing the open sixth E/C—though G is of course present in the backing vocals.

²⁵ Osborn (2017).



EXAMPLE 6. *The double-tonic complex plotted on Gottfried Weber's table of key relations. C minor and A major are both one step away from the A/C complex*

So why not just say the song is in A minor then? I must admit that I find a monotonal reading in A minor to be perfectly valid, and I cannot entirely disagree with it. But one thing that consistently undermines A minor is the presence of A $\flat$ s and B $\flat$ s in every section of the song. In the verses, the A $\flat$ - and B $\flat$ -major chords arise within the local C-major context as $\flat$ VI and $\flat$ VII. In the chorus, ostensibly centered on A, these notes are harder to explain; C-minor harmonies are out of place in an A-minor context. In a double-tonic complex, though, A minor and C major are both active at the same time, so the use of C minor is not foreign even to a seemingly A-focused passage. Example 6 plots the double-tonic complex on Gottfried Weber's table of key relations.²⁶ The complex takes up two nodes, and C minor is just one step away. The use of A $\flat$ s in an A-minor context is what we might call *double-tonic modal mixture*, the use of the parallel mode of one of the two tonics in the complex.²⁷ Furthermore, the chorus's concluding cadence uses A $\flat$ s and B $\flat$ s to lead into the four-note A–C–E–G sonority with semitonal voice leading. Measures 29 and 30 contain an A $\flat$ -major triad with a B $\flat$ bass note, one of the song's many soul-dominant sonorities. The bass's B $\flat$ descends by semitone to A, the C stays put, the E $\flat$ ascends by semitone to E, and the A $\flat$ descends by semitone to G (the same chord resolves to C major connecting the introduction to the first verse). Looking now at the end of the bridge, with its V⁷–i progression in A minor, we see that the E⁷ chord's G $\sharp$ s don't resolve up to A but in fact slide down to G $\flat$ s. Perhaps this voice leading refers back to the chorus's voice leading, in which A $\flat$ resolved to G. In other words, maybe this G $\sharp$ is functioning partly as an A $\flat$ when the E⁷ chord resolves to the four-note double-tonic chord—with both G and A stable members of the tonic sonority, G $\sharp$ /A $\flat$ is a tendency tone that pulls both upward and downward by semitone.²⁸

In sum, even though A-centricity seems to govern more than half of the song, some C-major sensibility is always present. Though one could read C major as a competing, “lurking” key threatening A's dominance, I find it more

convincing to see the two keys as forming a tonal partnership. When the bass and upper voices are stratified in the chorus, they still combine to form common chord types within Hall and Oates's soul-infused style.²⁹ The bass alone gives A-centricity, the upper voices alone give C-centricity, and both together form not discordant polytonality but a unified tonality blending A minor and C major. We can perceive tonic function on the A–C–E–G chords, and we can perceive the tension created by semitonally resolving tendency tones on the A $\flat$ /B $\flat$ chord in mm. 29–30. The absence of a single note or triad representing tonic stability does not negate the presence of such stability.

THE AEOLIAN-IONIAN COMPLEX

The relative key relationship is special. Not only are relative minor and major keys based on the same diatonic collection, but chords generally reside within the same functional category in both keys. For example, a D-minor chord is a subdominant harmony in both A minor (as iv) and C major (as ii). The similarity between the two keys is maximized when the minor member employs the Aeolian mode. If we were to take an Aeolian passage with A as tonal center and replace all the A-minor chords with C-major chords, the result would often be perfectly logical and functional in C major (and vice versa).³⁰ Consider the progression $\flat$ VI– $\flat$ VII–i, a common schema often referred to as the “Aeolian progression.”³¹ Mark Richards notes that Aeolian progressions often do not unequivocally project either $\flat$ VI– $\flat$ VII–i or IV–V–vi, creating at the least an ambiguity between two relative keys.³² Nicole Biamonte analyzes the Aeolian progression in relation to the standard major-mode IV–V–I: as shown in Example 7(a–b), both progressions express the functional sequence subdominant–dominant–tonic and are common cadential formulas.³³ Of course, IV and V in a major key are $\flat$ VI and $\flat$ VII of its relative minor, so if we take Example 7(b) and substitute an A-minor chord in the final slot, we get the Aeolian progression in A minor, as

29 Anyone unconvinced of the normality of the soul-dominant sonority need only listen to the duo's earlier single “She's Gone,” where the climax involves a series of truck-driver modulations created by sliding the soul dominant upward by semitone, from A/B to B $\flat$ /C to B/C $\sharp$ to C/D. See Spicer (2017, 2–4).

30 A great example of such tonic replacement, originally noted by Doll (2011, 4), occurs in Men At Work's “Down Under,” where the verse and chorus have identical chord progressions except for the first chord, which is Bm in the former and D in the latter. Changing this one chord shifts the centric emphasis from B to D, which Doll interprets as a modulation from B minor to D major. A similar example arises in Neil Young's “Rockin' in the Free World,” where the verse loops the chord progression Em–D–C and the chorus loops G–D–C–Em. As de Clercq argues, the emphasized G-major chord suggests the possibility of a modulation to the key of G major, but the chorus nevertheless remains somewhat “within the pull of the E-minor harmony” (2012, 51–52).

31 Biamonte (2010, 101–4); see also Doll (2017, 99–100).

32 Richards (2017b).

33 Biamonte (2010, 101).

26 Weber ([1832] 1851).

27 See also Lewis (1984, 6–8), for a similar idea.

28 Note that the chorus that follows the bridge begins with only the vocal parts, thus containing only the C-major portion of the stratified chorus texture.

a) C Aeolian b) C Major

S → D → T S → D → T

C: $\flat VI$ $\flat VII$ I C: IV V I

c) A Aeolian d) Double tonic

S → D → T S → D → T

a: $\flat VI$ $\flat VII$ i { C: IV V I
a: $\flat VI$ $\flat VII$ i

EXAMPLE 7. *The Aeolian progression's kinship with IV–V–I allows the possibility of combining the two, forming a double-tonic complex (a–b reproduced from Biamonte [2010])*

shown in Example 7(c). Now, what if we combine Examples 7(a) and 7(b) by keeping the top staff as a C-major chord but using an A in the bass, as in Example 7(d)? Might that chord in some situations be heard not as $A\text{m}^7$ —that is, as an A-minor triad colored with another note—but instead as both an A-minor and a C-major triad at the same time? We would still perceive the functional progression subdominant–dominant–tonic, but the Roman-numeral distinction of $\flat VI$ – $\flat VII$ –i versus IV–V–I would remain unspecified. The result would suggest a double-tonic complex.

Example 8 transcribes the chorus from LMFAO's 2011 electropop hit "Party Rock Anthem." The accompaniment's groove, which underlies most of the song, presents the same progression from Example 7(d) in reverse order, transposed to begin on an apparent $F\text{m}^7$ chord. The melodic line seems indifferent to the bass's emphasis on F, outlining an $A\flat$ -major triad with a prominent 3–2–1 descent across each four-bar phrase. The four-note chord F – $A\flat$ – C – $E\flat$ can be understood to function as the union of F-minor and $A\flat$ -major triads (hence the example's notation of " $A\flat/F$ " rather than " $F\text{m}^7$ "), representing the four-note tonic of the $F/A\flat$ double-tonic complex. As with "Private Eyes," the bass and upper voices are tonally stratified such that the former focuses on F and the latter focus on $A\flat$. Though I acknowledge the possibility of hearing the song in $A\flat$ major, with the loop in Example 8 interpreted as vi – V – IV , the parts of the song that do *not* contain that loop emphasize F, especially the instrumental post-chorus at 0:58.³⁴ I find it more convincing to treat this song's chorus as a fusion of the Aeolian and Ionian modes, with the $A\flat/F$

chord functioning as the tonic of both, and the $E\flat$ and $D\flat$ chords acting respectively as dominant and subdominant harmonies within both orientations; in other words, the progression combines I–V–IV in $A\flat$ major and i – $\flat VII$ – $\flat VI$ in F minor.

Tina Turner's "What's Love Got to Do with It?" at first establishes a clear $G\sharp$ -minor tonality with a bass pedal lasting from the introduction through the first verse. The ensuing pre-chorus shuttles between E- and $F\sharp$ -major triads, suggesting the Aeolian progression's $\flat VI$ and $\flat VII$. Example 9 shows the ensuing transition from prechorus to chorus; as the example shows, the downbeat of the chorus does not provide an unequivocal i chord completing the projected cadential return to a $G\sharp$ -minor tonic. Instead, the accompaniment drops out for the first eighth note (except for a soft electric piano $G\sharp$ in unison with Turner's melodic note); when it returns, the bass jumps in on B and the guitar provides a $B/D\sharp$ dyad with its reggae-infused offbeats. The keyboard does play a $G\sharp$ -minor triad, though, so the most "correct" interpretation is probably that this chord is i^6 in $G\sharp$ minor, with the bass simply providing a melodic embellishment of tonic. However, something about that moment gives more than a hint of a B-major harmony. The textural break withholds any support for the melodic $G\sharp$, and the bass and guitar's strong entrance on the second eighth note implies a B-major triad, even with the fifth omitted. (The keyboard's $G\sharp$ -minor triad, on the other hand, is very quiet in the mix, and enters after the bass's and guitar's chord.) The B-major possibility strengthens in the chorus's subsequent phrases, where there is no $G\sharp$ on the downbeat, so the first thing we hear is the $B/D\sharp$ dyad in the bass and guitar.³⁵

Hearing B major at the chorus reflects a generous serving of the aforementioned "major-mode bias." In "What's Love Got to Do with It?," there is ample evidence supporting a $G\sharp$ -minor tonic, while the support for B major is essentially limited to the chorus's first bass note; nevertheless, the support for $G\sharp$ minor is not so unequivocal as to negate the major-mode bias. Depending on your personal level of major-mode bias, you may hear the chorus as solidly in $G\sharp$ minor, solidly in B major, or anywhere in between. Listeners closer to the former category, though, may have their hearings complicated later in the song. After the second chorus, a solo section begins over the chorus's accompaniment, shown in Example 10, but here the vocal melody's opening $G\sharp$ is replaced by the synthesized harmonica's sustained $F\sharp$.³⁶ The result is a complete B-major triad on the downbeat, which combines with the electric

serve as a dance break. The pop-drop derives from EDM forms (see Butler [2006]).

34 This post-chorus acts as what pop critic Charlie Harding (2016) calls the "pop-drop": a climactic instrumental section placed after the chorus to

35 I am not the only one who hears a major chord here; the McGill Billboard Corpus, a collection of manually annotated rock songs, analyzes these chords as B chords; see <http://ddmal.music.mcgill.ca/research/billboard>.

36 This moment coincides with a pump-up modulation such that the rest of the song is transposed up a whole step; for ease of reading, my transcriptions and discussion are written as if there were no modulation.

CHORUS

at 0:29

Melody outlines
A♭ major

(3̂ 2̂ 1̂)

A♭/F E♭ D♭ A♭/F E♭ D♭

Party rock is in the house tonight. Everybody just have a good time.
And we gon' make you lose your mind.

$\left\{ \begin{array}{l} \text{A}\flat: \text{I} \\ \text{f: i} \end{array} \right.$
 $\begin{array}{cc} \text{V} & \text{IV} \\ \text{♭VII} & \text{♭VI} \end{array}$
 $\left\{ \begin{array}{l} \text{I} \\ \text{i} \end{array} \right.$
 $\begin{array}{cc} \text{V} & \text{IV} \\ \text{♭VII} & \text{♭VI} \end{array}$

T D S T D S

EXAMPLE 8. LMFAO, “Party Rock Anthem” (2011): combination of I–V–IV and i–♭VII–♭VI, where the tonic chord is a four-note union of F-minor and A♭-major triads

piano’s G♯-minor triad on the second beat to form the four-note sonority B–D♯–F♯–G♯. The solo line itself seems firmly grounded in B major, giving a palpable sense of I–V–IV in the accompanying chord progression, rather than i–♭VII–♭VI. After this point, there are no more verses (which offered the strongest support for a minor tonic), and the ensuing bridge, shown in Example 11, leans further toward B major; the bridge’s final B chord leads right into the return of the chorus, highlighting the chorus’s B-major possibility. Interpreting a double-tonic complex to underlie “What’s Love Got to Do with It?” allows the minor and major tonics to coexist within a cohesive tonal system. The verses explore the minor side of the complex, the bridge explores the major side, and the prechorus and chorus combine the two to varying degrees throughout the song.

I mentioned above that the Aeolian and Ionian modes are often indistinguishable beyond their different tonic chords; “Party Rock Anthem” and “What’s Love Got to Do with It?” exhibit a tonal situation where the two tonic chords are both present within a single four-note sonority. There is no tonal conflict or competition between two keys; instead, the songs are content not to specify a tonal center or single governing tonic triad. The result is a specific type of double-tonic complex we might call the “Aeolian-Ionian complex,” where a diatonic collection is organized to project either an Aeolian or Ionian tonic without settling on one or the other. In “Party Rock Anthem,” hypermetrical emphasis on the A♭/F chords solidifies their status as tonic, with the E♭ and D♭ chords effecting a “retrogressive” T–D–S fall away from that tonic. In “What’s Love Got to Do with It?,” the prechorus’s E and F♯ chords project a “progressive” S–D–T resolution into the chorus’s opening chord. That the tonic chords combine two triads with two different roots does not take anything away from their tonic-ness. That is, we have the makings of a structural

double-tonic complex, where a four-note chord, as a union of a minor and a major triad, acts as both a stable surface sonority and a generative tonic. Traditional tonal relationships do not disappear in the context of this four-note tonic: melodic tendency still gravitates stepwise toward tonic members as in Example 12(a), and chords still divide into three functional categories as in Example 12(b). (Example 12[b] adapts Biamonte’s functional diagrams; the chords in parentheses do not usually occur in double-tonic contexts.) In other words, the four-note tonic is not just a surface sonority but rather represents the “harmonic embodiment” of the double-tonic complex “at work throughout the structure.”³⁷

THE AXIS PROGRESSION AND “BUILDING A MYSTERY”

One might wonder if double-tonic complexes, specifically those of the Aeolian-Ionian variety, can arise in the absence of four-note tonic sonorities appearing on the surface. Despite the prevalence of seventh chords throughout the repertoire, rock’s chord palette remains largely built on triads.³⁸ I believe that many of rock’s triadic progressions can be understood in reference to a double-tonic complex. In particular, songs that use only the five non-parenthesized triads from Example 12(b) with emphasis on both Am and C, but stronger emphasis on the former, seem especially suited to a double-tonic interpretation. Rock’s “axis progression”—Am–F–C–G and its transpositions, usually presented as a repeated chord loop—is a famously ambiguous progression that fulfills all of these criteria. As Richards discusses in detail, some instances of this progression are clearly in A minor, some others are clearly in C major, but many—perhaps most—do not firmly establish

³⁷ Bailey (1985, 122).

³⁸ Temperley (2018, 42–43).

end of prechorus... **CHORUS**

at 0:46

E F# G#m/B F# E F#

You must try to ignore that it means more than that, whoa. What's love got to do, got to do with it?

e. piano e. guitar

bass

Bass and guitar hint at a B chord, while e. piano plays G#m

G#m/B F# E F# G#m/B F# E F#

What's love but a secondhand e - mo - tion? What's love got to do, got to do with it?

After the first phrase, there is no G# on the downbeat

G#m/B F# E F#

Who needs a heart when a heart can be broken?

EXAMPLE 9. Tina Turner, "What's Love Got to Do with It?" (1984), prechorus into chorus: the chorus's opening tonic contains the notes of G# minor but the bass and instrumental texture hint at a B-major possibility

either key unequivocally.³⁹ Doll interprets the third option as a series of rapid reorientations from hearing one center to the other: "With this kind of . . . ambiguity we get a consistent oscillation from one center up three semitones to another center, often between 'relative' minor and major tonic chords from

Aeolian and Ionian orientations of the same white-key diatonic pitch-class set."⁴⁰ Likewise, Richards asserts that in "ambiguous mixes" of the two tonalities, one's tonal interpretation "may shift in the midst of hearing the chorus, upon hearing repetitions of the chorus, or upon repeated listenings to the song."⁴¹

³⁹ Richards (2017a) employs the term "Axis progression" to refer to all rotations of the progression, not just those beginning on the A-minor chord; to distinguish, he refers to this specific rotation as "Axis-a."

⁴⁰ Doll (2017, 252).

⁴¹ Richards (2017a, 29).

SOLO

at 2:02 G#m/B F# E F# G#m/B F# E F#

The solo line completes a B-major triad on the first beat, forming the four-note sonority B-D#-F#-G# on the second beat

EXAMPLE 10. Tina Turner, “What’s Love Got to Do with It?,” beginning of solo section (transcribed a whole step below the sounding pitch to facilitate comparison with previous example): the solo line strengthens the B-major possibility

Doll’s and Richards’s interpretations stem from the assumption that at any given moment we are perceptually oriented toward a single specific tonal center, even if we quickly reorient to a different one. In the Axis progression, I am not so sure that assumption is always warranted. When I listen to the Axis progression, I do not necessarily hear it only in A or only in C, but neither do I feel tossed around on a tonal rollercoaster, bouncing back and forth between the two. Rather, the progression’s first and third chords *both* seem to project tonic stability, resulting in a constant ebb and flow between stable (Am and C) and unstable (F and G) sonorities. [Example 13](#) shows how the voice leading works. Here, stable tones are shown with open noteheads, and active tones are shown with flags. We can see that even without a single specified tonal center, the progression outlines the succession from tonic to subdominant to tonic to dominant. If we define subdominant function by the presence of a downward-pointing $\hat{6}$, as do both Doll and Biamonte, then the F chord qualifies in both A and C. Similarly, if dominants arise from $\hat{7}$ and $\hat{2}$, the G chord again qualifies in both keys. The two tonic chords contain the stable tones C and E, with the third note oscillating between A and G. Both A and G first arrive as stable tonic tones but become activated when the chord changes to a subdominant or dominant. In this way, although all four chords are triads, we could understand the progression to compose out the four-note tonic A–C–E–G, representing the union of A-minor and C-major triads.

One of music theorists’ favorite examples of the Axis progression is Sarah McLachlan’s “Building a Mystery.” As shown in [Example 14](#), the song loops the progression, here beginning on B minor, in its verses and choruses, separated by a different progression in the prechorus. [Example 15](#) lists some previous

interpretations of this song’s tonality. Koozin focuses on the competition between the two potential tonics, while Doll and Richards hear fluctuations between them. Attas and de Clercq are more concerned with form, but both acknowledge two possible tonal interpretations. Osborn comes down in favor of a monotonal reading in D major, arguing against B minor as a possible tonic due to the lack of its V chord (i.e., a chord built on F#). Despite our various tonal interpretations, we all agree that determining this song’s key is not a trivial matter. What I am suggesting with a double-tonic reading is that the lack of an obvious single key doesn’t necessarily indicate a musical *problem* in need of an analytical solution. Rather, we can hear the Axis progression as a stable chord loop prolonging tonic function even in the absence of a specific tonal center. When the B-minor chord arrives at the beginning of the chorus, it has the stability of a tonic chord; two chords later, the D-major chord *also* acts as a stable tonic chord, rhetorically emphasized with the title of the song in the lyrics over a melodic arpeggiation of a D-major triad.

If the verse and the chorus prolong the double-tonic sonority B–D–F#–A, as demonstrated, the prechorus connects them with subdominant and dominant sonorities within this double-tonic context. The section’s progression, involving a shuttle between E-major and G-major chords followed by A major, works perfectly well in either B Aeolian or D major; II#–IV–V and IV–bVI–bVII are both common rock progressions leading to tonic (e.g., the Beatles’ “Eight Days a Week,” end of bridge, for the former, and Bob Seger’s “Turn the Page,” end of chorus, for the latter). In both cases, the E and G chords group together as a subdominant prolongation, with the A chord serving as dominant. A harmonic representation of the verse–prechorus–chorus cycle might look like [Example 16](#),

BRIDGE

at 2:21 A/B B A/B B

I've been taking on a new di-rec-tion but I have to say

I've been thinking 'bout my own pro-tec-tion; it scares me to feel this way. Hey,

The bridge begins with the song's first significant emphasis on B major

The bridge ends on a B chord, though it sounds a bit like V of E major

CHORUS

G#m/B F# E F#

What's love got to do, got to do with it?

etc.

When the chorus returns immediately following the bridge's final B chord, the first chord's B-major possibility is especially potent

EXAMPLE II. Tina Turner, "What's Love Got to Do with It?," bridge (transcribed a whole step below the sounding pitch): this section moves away from G# minor, and when it leads back to the chorus it strongly projects B major

with composed-out four-note chords in the tonic and pre-dominant areas.⁴² This harmonic layout—tonic prolonged in the verse, pre-dominant to dominant in the prechorus, and a

return to tonic at the onset of the chorus—is the standard functional layout in verse-prechorus-chorus form.⁴³ The

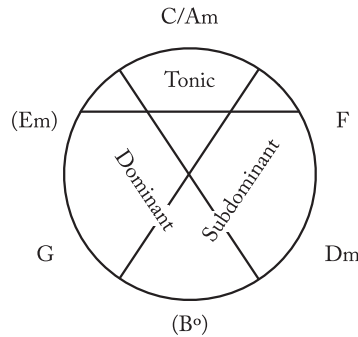
⁴² I differentiate between the *subdominant*, referring to a category of chords defined in reference to the IV chord, and the *pre-dominant*, referring to a syntactical function appearing between an initial tonic and a dominant within a large-scale harmonic trajectory. See Nobile (2016; 2020,

Chapter 1). In this case, as is common, pre-dominant function is carried by chords in the subdominant category.
⁴³ See Nobile (2020, Chapter 8).

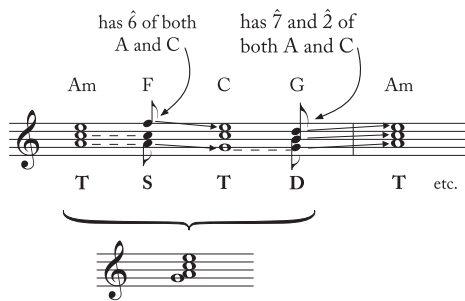
- a) Non-tonic notes gravitate stepwise toward tonic notes; G and A, both members of the four-note tonic sonority, can move stepwise to one another.



- b) Tonic, subdominant, and dominant functional categories are explicitly defined for diatonic triads. Songs exhibiting an Aeolian-Ionian double-tonic complex do not usually contain the chords in parentheses. This diagram adapts Nicole Biamonte's functional diagrams (Biamonte [2010]), which in turn adapt Eytan Agmon's diagrams (Agmon [1995]).



EXAMPLE 12. *Melodic tendencies and harmonic categories within a double-tonic complex*



EXAMPLE 13. *The Axis progression seen as prolonging a four-note sonority with stepwise voice leading*

B-minor chords are tonics, and the D-major chords are tonics, and the song's tonality is the B/D complex.⁴⁴

44 One may wonder about a system for labeling double-tonic progressions with Roman numerals or some other transpositionally invariant system. Theorists have already encountered this problem with the Axis progression, which needed a specific name because we could not call it either "vi-IV-I-V" or "i-♭VI-♭III-♭VII" without specifying a tonal center. Within a diatonic collection, one could use Riemannian function labels, specifying major or minor versions, such as T^M and T_m for C major and A minor respectively. Alternatively, though perhaps too confusingly, one could resort to a sort of "la-based minor" system, where "do" and the Roman numeral "I" apply to what would be the Ionian tonic (i.e., C in the white-key collection), regardless of the song's tonal center (or lack thereof). Thus, songs in Aeolian would be centered on la and vi chords would be tonics; songs in Dorian would have re and ii as tonic, etc. The Axis progression would be labeled vi-IV-I-V with the understanding that those numerals do not point to any single chord as tonic; double-tonic songs in general would have either vi and I as possible tonics (plus their union—"I/vi"?).

THE ABBEY ROAD MEDLEY

Before concluding, I would like to acknowledge the possibility that rock music can exhibit larger-scale double-tonic complexes more akin to those seen in Bailey's nineteenth-century examples, where both component keys exist independently over the course of a multi-part work, coming together at important structural moments. A couple of theorists have pointed out the tonal similarities between *Tristan* and the medley from side two of the Beatles' album *Abbey Road*. Both pair passages centered on A with others centered on C; the *Abbey Road* medley does so across eight tracks spanning over sixteen minutes of music. Gauldin posited a double-tonic reading for the *Abbey Road* medley in 1990, and Everett picks up on the idea of the A/C pairing as a "central concern" in his detailed analyses.⁴⁵ Though he cites Gauldin's double-tonic reading, Everett's analysis leans more toward tonal pairing, specifically via *associative tonality*, another Baileyan concept where each key has a specific narrative significance. To Everett, the two keys are juxtaposed to highlight an opposition in the text: "A contains tension, and C repose. . . . Much of the poetic text of the medley deals with selfishness and self-gratification. . . . These selfish moments are all set in the context of the tonal center of A," while moments underscoring generosity "are the points where C major is central."⁴⁶ Even Gauldin hints that he might be thinking more along the lines of tonal pairing than a unified double-tonic complex: "what I find most

45 Everett (1995; 1999, 260–71).

46 Everett (1999, 271).

VERSE

0:12

Tonic

Bm G D A Bm G D A etc.

You come out at night That's when the energy comes, and the dark side's light, and the vampires roam

PRECHORUS

1:11

Subdominant

E G E G A

You're so beautiful with an edge (and) charm and, and so careful when I'm in your arms

CHORUS

1:23

Tonic

Bm G D A Bm G D A etc.

'cause you're workin', building a mys - tery Hold - ing on and hold - ing it in

EXAMPLE 14. Sarah McLachlan, "Building a Mystery" (1997)

intriguing is the tonal *conflict* created by the *struggle* for control between the two chief protagonists of A and C.⁴⁷

The relevant question to this article is, could we understand the entire medley, parts of which are definitely in A and other parts of which are definitely in C, to exhibit a large-scale double-tonic complex where A and C are *unified*, not opposed? I believe that certain aspects of the medley suggest that the answer is yes. Example 17 gives a skeleton reduction of the entire medley. The opening track, "You Never Give Me Your Money," presents four types of tonic. The first sound we hear is the sonority A–C–E–G, here acting as the tonic of A minor prolonged via the circle-of-fifths sequence shown in Example 18(a). After three iterations of the sequence, we arrive at a contrasting theme where C major is tonic, shown in Example 18(b), again involving a sequential progression. Notice that both sequences tonicize the other member of the A/C complex along the way. Example 18(c) summarizes the final portion of the track, which centers on A major and ends by shuttling back and forth between C- and A-major chords. The quick alternation of C- and A-major triads can be understood as an unfolded expression of the four-note double-tonic sonority, with double-tonic modal mixture creating the chromatic shift from C to C $\sharp$ (recall Example 6). An A is actually present in the melodic material over many C chords (as in Example 18[c]); the same is true of G over A chords. Everett explicitly relates this passage to the opening four-note sonority: "Harrison's voice freezes G $\sharp$ as a pentatonic member of both chords; the G never really resolves, but remains above the A harmony. In fact, the C chord is heard as its consonant support. The song's opening chord is, of course, A–C–E–G, and

as in 'A Hard Day's Night,' the piece ends with a dissonant sonority that relates to the work's opening."⁴⁸

The four tonic chords in the medley's opening track suggest that A and C are more entangled than is necessarily expected in a case of tonal pairing. Both keys are certainly present individually, but neither has the chance to firmly establish itself as independent from the other. As the medley goes on, the entanglement deepens. The second portion of the melody, comprising "Sun King" through "She Came in Through the Bathroom Window," develops the tonal framework with excursions to subordinate key areas. "Sun King" begins and ends in E major, suggesting a move to A major's dominant area. However, an abrupt shift after the initial E-major groove brings us to an F/G chord, a soul dominant that sends us into C major for the track's main theme. The lyrics "here comes the sun," of course, refer back to George Harrison's song that opened the album's second side; when we get the lyrical surprise "... king!," the harmony opens up to A major, now reminding us of the C-major/A-major pairing from the end of "You Never Give Me Your Money." A major is here fleeting and subordinate to the local tonic of C major. When the key of E major returns at the end of "Sun King," it holds on as we move to the next track of the medley, "Mean Mr. Mustard." "Mean Mr. Mustard" and "Polythene Pam" both remain grounded on E, the latter track adding D $\sharp$ (via the double-plagal chord loop D–A–E), ultimately turning E major into V⁷ of A. The final track of the medley's second part, "She Came in Through the Bathroom Window," is in A major with an internal excursion to C major ("Sunday's on the phone to Monday..."), reversing the hierarchical relationship seen in

47 Gauldin (1990, 151, emphasis mine).

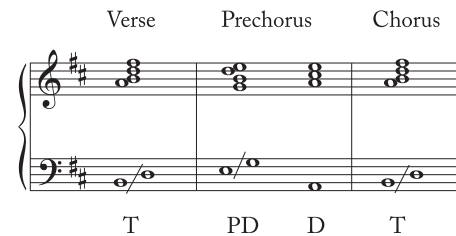
48 Everett (1999, 263).

Koozin (2008)	D major with “competing modal/ linear pattern around B minor”
Doll (2011)	Verse is in B minor (although D “threatens to break through”); choruses are in D
Richards (2017a)	Tonal center changes from phrase to phrase in verses; choruses are in D
Attas (2011)	“Circular chord progression ... does not make a particular tonality clear”
de Clercq (2012, 96–98)	D major, with B minor an acknowledged alternative
Osborn (2017, 67)	D major

EXAMPLE 15. *Previous tonal interpretations of “Building a Mystery”*

“Sun King.” The track ends with yet another C-major–A-major oscillation.

The medley’s third and final portion begins with “Golden Slumbers” and the return of the A–C–E–G sonority shown in [Example 19](#) (E is literally absent, but implied). As in “You Never Give Me Your Money,” this chord kicks off a circle-of-fifths sequence, but here it does not make it all the way back to A, instead settling on C. The potential for the four-note sonority to function as a union of two triads is stronger here than at the beginning of the medley. Paul McCartney’s piano voicing places the C–G fifth on strong beats, and his vocal line emphasizes G, suggesting perhaps an inverted C⁶ sonority. The loud bass note A that enters after a few measures, on the other hand, emphasizes the sonority’s A-minor potential. A minor quickly gives way to C major, though, and especially with the arrival on C accompanying the word “home,” we might think that the medley has settled on C as its global tonic. “Carry that Weight” adds to the C-major emphasis, but just as we might be forgetting about the hint of A minor that opened “Golden Slumbers,” the A–C–E–G sonority reappears in climactic fashion. The sonority’s return coincides with the recapitulation of the medley’s opening track, “You Never Give Me Your Money,” first instrumentally with a brass ensemble and eventually with a new set of lyrics. The medley’s opening theme returns here under circumstances somewhat different from its initial presentation; with C-major themes surrounding it, and after the fused triads of “Golden Slumbers,” its previous status as projecting an A-minor tonic is called into question. In addition, any sense of *progressive* tonality,



EXAMPLE 16. *Harmonic reduction of “Building a Mystery” showing prolonged four-note chords giving rise to a double-tonic complex*

where A gives way to C over time, is countered by the reemergence, untransposed, of the opening material.

This moment underscores the interconnectedness of A and C throughout the medley. From the beginning, neither center has been able to establish itself without reference to the other. As both Everett and Gauldin note, the medley gradually shifts from favoring A to favoring C; specifically, the first half, comprising [Example 17’s](#) top staff, tends to treat C major as an interior key within a broader A-minor or A-major context, and the second half, comprising [Example 17’s](#) lower staff, more often places A-centric passages within C-major contexts. Faced with this tonal situation, our instinct as analysts might be to consider A and C as dramatic agents competing for supremacy, as in Gauldin’s “struggle for control between the two chief protagonists of A and C.” Such a reading, valid as it may be, nevertheless imposes a monotonal bias: only if we assume two keys cannot coexist do we arrive at a narrative of competition. To me, the music itself does not project any sort of tonal struggle; harmonic tension, insofar as it exists, might be better attributed to a conflict between C₄ and C₅ than to the tonal centers of A and C. The double-tonic complex provides an alternative to the conflict paradigm. That is, it allows us to consider the possibility that A and C are not opposing tonal forces but rather two sides of the same coin, an “irreducible balance,” in BaileyShea’s words.⁴⁹ In this interpretation, all of the medley’s tonics can be understood to derive from the four-note sonority A–C–E–G: C major is a partial statement, A major is a partial statement with a chromatic alteration, the alternating C- and A-major triads are a linearization of the four-note complex, and of course the A–C–E–G chords occurring at the beginning of the medley’s first and third parts and at its climax are complete statements of the prolonged tonic sonority.

Reading a double-tonic complex rather than two competing keys carries implications for any expressive interpretations that might follow. In Everett’s analysis, the medley’s final track, aptly titled “The End,” provides a summation of its tonal and narrative arcs. The track takes us from A major to C major over the lyric “and in the end, the love you take is equal to the love you make.” This line, to Everett, signifies the realization that “there is only as much self-gratifying love (‘the love you take’), that of A major, as there is of the generous kind (‘the

49 BaileyShea (2007, 193).

Annotations in the score include:

- A-C-E-G
- C-major triad
- A-major triad
- V/A → V/C
- four different kinds of tonic within DTC
- A-C-E-G
- climactic statement of A-C-E-G sonority

EXAMPLE 17. *The Beatles, medley from Abbey Road (1969), reduction. As in Bailey's nineteenth-century examples, both C and A are central for portions of the medley, but neither emerges as the global tonic*

love you make'), that of C major."⁵⁰ Reading A and C as competing, mutually exclusive keys suggests a narrative of generosity triumphing over selfishness, a narrative to which Everett alludes with his graph showing C major as the medley's primary key. If instead we read A and C as two sides of a unified whole, the narrative becomes one of interconnectedness. The love you take and the love you make are not independent but mutually reinforcing; you can't have one without the other. The framework of a double-tonic complex embodies this narrative fusion: neither A nor C exists on its own, but each only as one side of a broader, more abstract tonality.

CONCLUSION: THE DOUBLE-TONIC COMPLEX AND COGNITIVE ENGAGEMENT

Cognitive scientists—cognitive linguists in particular—talk about cognitive engagement at a specific conceptual level. For instance, the sentences "I have five pets," "I have five dogs," and "I have five Weimaraners" engage us at different levels of meaning specificity. Example 20(a) uses David Tuggy's graphical system to model these different levels.⁵¹ In the example, conceptual levels are displayed vertically, with higher nodes more abstract than lower nodes. Thicker borders around a node signify deeper cognitive engagement at that particular level.⁵² Importantly, the sentence "I have five pets" is *not* ambiguous, even though it has multiple mutually exclusive

subcases, nor does the word "pets" evoke multiple simultaneous concepts; it does not signify dogs and cats at the same time, but rather signifies a single, more abstract concept of which dogs and cats are subsets. Linguists refer to such situations variously as *indeterminacy*, *vagueness*, or *generality*. Linguistic *ambiguity*, on the other hand, arises when there are multiple mutually exclusive possibilities within the conceptual level at which we cognitively engage. As diagrammed in Example 20(b), ambiguity refers to situations where the possibilities cannot be combined into a single more general meaning. For instance, if I say "I am arriving at the bank," I could be referring to a riverbank or a financial institution. The two possible meanings of "bank" are not reconcilable into a general meaning encompassing both possibilities. Furthermore, as linguist Alan Cruse puts it, "the interpretation cannot be left undecided: both speaker and hearer must select a reading (the same reading) if the sentence is to play its part in a normal conversational exchange."⁵³ In other words, linguistic ambiguity is a semantic problem in need of a solution.⁵⁴

Most discussions of tonality assume that we always cognitively engage at the level of key. With this assumption, passages that do not specify a single tonic are ambiguous, just like the sentence with "bank." Tonal ambiguity means conflict, competition, or confusion. Example 21(a) demonstrates tonal ambiguity by adapting the linguistic diagram from

while exiting a ballpark concession stand might evoke an entirely different concept from that in Example 13.

⁵⁰ Everett (1999, 271).

⁵¹ Everett (1999, 271).

⁵² Tuggy (1993).

⁵³ The contents of the nodes are the *concepts* suggested by, but distinct from, the words we hear. For instance, the sentence "I have five dogs" spoken

⁵⁴ Cruse (1986, 51).

For more on the linguistic concepts discussed in this paragraph, see Cruse (1986, Chapter 3), Langacker (1987), Tuggy (1993), and Klein and Murphy (2002).

a) “You never give me your money” theme (0:22): Am7 chord as tonic

You never give me your money You only give me your funny pa - per And in the middle of negotiations, you break down.

circle-of-fifths sequence in *A* minor

C is tonicized within sequence

b) “Out of college” theme (1:08): CM chord as tonic

Out of college, money spent See no fu - ture, pay no rent All the money's gone, nowhere to go

descending-thirds sequence in *C* major

A is tonicized within sequence

c) “One sweet dream” theme (2:27): AM chord as tonic

One sweet dream

A in melody over CM triad

came true today, yes it did! Na guitar

(2:57): alternation of CM and AM chords

EXAMPLE 18. *Four types of tonic in “You Never Give Me Your Money”*

vocal

Once there was a way

piano

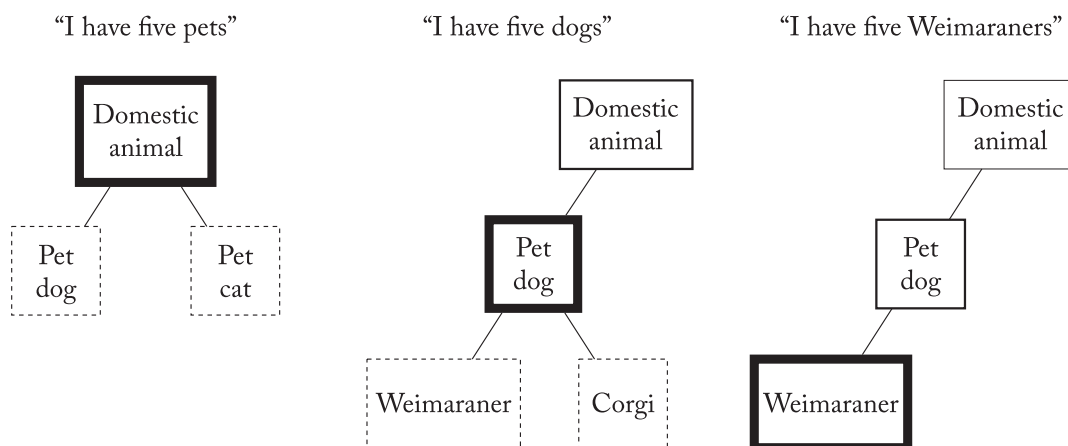
bass

EXAMPLE 19. *Opening of “Golden Slumbers”*

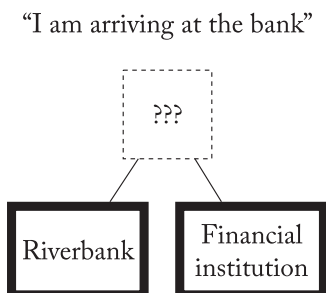
Example 20(b). This model represents a musical situation where two possible keys compete for our attention. We feel cognitive tension because we are being asked to choose between two equally plausible but mutually incompatible hearings, just as we would upon hearing the sentence with “bank” without any clarifying context. (However, unlike most cases of

linguistic ambiguity, tension resulting from tonal ambiguity is often fully intended.) Here, Doll’s description of tonal ambiguity as “facing two different directions at once” or “flickering” between two incompatible interpretations is apt. What I have proposed in this article is the possibility of a tonal situation more akin to linguistic *indeterminacy*, as shown in Example 21(b) (compare Example 20[a]). In this model, our cognitive engagement is *not* at the level of key, but instead jumps up a level to where tonality encompasses multiple keys within a single concept. As I hope to have shown, two relative keys in the rock idiom can be so intertwined so as to blend together, their differences reduced to small details that fade from our mental focus. The double-tonic complex is to its two component keys as pet is to dog and cat. Harmonic function, syntactical patterns of tension and resolution, and other normal tonal relationships are still there; all that has been removed is a specific pitch-class center.

a) Linguistic indeterminacy: “pets” does not signify dogs *and* cats but rather a single abstract concept

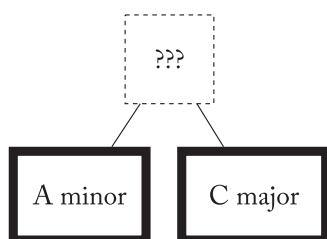


b) Linguistic ambiguity: two potential meanings do not combine into a single abstract concept

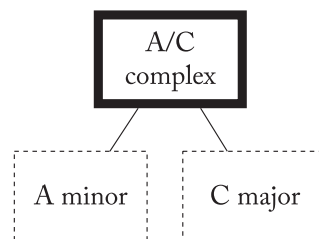


EXAMPLE 20. *Cognitive engagement occurs at different conceptual levels depending on the word used in a given sentence*

a) Tonal ambiguity



b) Double-tonic complex



EXAMPLE 21. *Tonal ambiguity, like linguistic ambiguity, involves two potential keys that are mutually exclusive. The double-tonic complex is analogous instead to linguistic indeterminacy, where cognitive engagement occurs at a more abstract conceptual level*

Relative key pairings are special in this way. A chord that is tense in one key generally remains tense in its relative key; as stated earlier, dominants and subdominants generally remain dominants and subdominants, and so on, especially when we consider the Aeolian mode as the default for a minor tonic. Other key pairings do not share this feature. Situations such as the oft-cited D/G ambiguity in Lynnyrd Skynyrd’s “Sweet Home Alabama” are therefore not easily explained through an

extension of the double-tonic framework; non-relative pairings would seem to give rise to a diagram like [Example 21\(a\)](#) rather than [Example 21\(b\)](#).⁵⁵ However, the idea that two non-relative keys could coexist without causing conflict, struggle, or

⁵⁵ Discussions of “Sweet Home Alabama”’s tonal ambiguity appear in [Doll \(2017, 222–38\)](#), [Tagg \(2014, 427–30\)](#), [Biamonte \(2010, 98–99\)](#), [Stephenson \(2002, 45–46\)](#), and many others.

ambiguity is intriguing, and one that would certainly reward further study. Finally, I should make clear that I am not trying to make the case that relative keys are *never* truly ambiguous. Rock is full of passages that exploit the tension created by two opposed key centers, often taking advantage of the expressive significance of minor/major conflict. Osborn's aforementioned analysis of Radiohead's "Reckoner," for instance, convincingly argues for a true competition between G major and E minor;⁵⁶ other songs that move between relative keys often do so to emphasize an opposition in the text, such as Neil Young's "Rockin' in the Free World." You might even think that one of the songs analyzed in this article is truly ambiguous, the two keys butting heads instead of fusing into a double-tonic complex. What I am advocating is the *availability* of a double-tonic analysis. In other words, when faced with two possible key centers, we should ask whether the musical situation warrants reading a tonal struggle, or whether it might simply exhibit a type of tonality where a specific tonal center is not necessary.

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